User Fairness Aware Power Allocation for NOMA-Assisted Video Transmission With Adaptive Quality Adjustment

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***Abstract*—In this paper, we investigate the user fairness power allocation (PA) for video transmission in a non-orthogonal multiple access (NOMA)-assisted wireless system with a pair of users. Considering that video transmission is usually delay-sensitive, the video quality can be adaptively adjusted based on the network conditions. First, a new performance metric, called the average video quality degradation probability, is proposed to describe the adaptive video quality. Then, based on the principles of NOMA and user queue model, the average video quality degradation probability of both users is derived. Next, a bisection search based power allocation (PA) algorithm is proposed to balance the video quality between the NOMA pair. Simulation results demonstrate that the proposed NOMA scheme is more effective to achieve the user fairness compared with the fixed NOMA scheme, the fractional transmit power allocation (FTPA) based NOMA scheme, the achievable rate fairness (ARF) NOMA scheme, and the optimized time division multiple access (TDMA) scheme.**

***Index Terms*—Video quality, non-orthogonal multiple access, user fairness, power allocation.**

I. Introduction

In recent years, video streaming transmission has been the main contributor to the data load of Internet of Thing (IoT) networks. The rapid increasing demand of multimedia video promotes the emergence of new delay-sensitive applications, such as ultra high definition (HD) video, cloud game, and virtual reality (VR). For these applications, the millisecond level delay is usually required [1]. Therefore, it is a big challenge to provide strict delay guarantee to video streaming transmission.

Considering the random characters of wireless transmission, we introduce the concept of adaptive video transmission. Specifically, the quality of real-time video streaming can be adaptively adjusted based on the network configurations to avoid unacceptable delay [2], [3]. However, when the selected quality is lower than a threshold, the experience of users will be also degraded. Therefore, it is important to investigate how to improve the quality for video streaming delivery with adaptive quality adjustment, especially in the spectrum constraint scenarios. Non-orthogonal multiple access (NOMA) has been proved to be an effective technology to improve the spectral efficiency (SE) [4], [5]. In NOMA-assisted networks, multiple users can share the same spectrum resource simultaneously with different allocated power.

Manuscript received February 28, 2021; revised June 22, 2021 and September 9, 2021; accepted October 24, 2021. Date of publication November 23, 2021; date of current version January 20, 2022. The work of Sangsha Fang and Pingzhi Fan was supported in part by the National Natural Science Foundation of China under Grant 61731017 and in part by 111 Project under Grant 111-2-14. The review of this article was coordinated by Prof. Xianbin Wang. *(Corresponding author: Sangsha Fang.)*

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Digital Object Identifier 10.1109/TVT.2021.3129805

Resulting from high SE, NOMA is adopted to improve the performance of video transmission [6]–[8]. However, the NOMA assisted adaptive video transmission has not been well investigated.

In the power domain (PD) NOMA systems, there exist conflicts between different users. Specifically, when a user is allocated more power, the performance of this user can be improved. However, other users will suffer performance degradation due to available power decrease. Therefore, it is impossible to maximize the video quality of all users in NOMA assisted video transmission systems at the same time. To maximize the sum of users' video quality, the user with strong link will have good quality, while the quality requirement of the user with weak link cannot be guaranteed. Since the quality of links are usually determined by the distances between transmitters and users. The phenomenon is called as the near-far unfairness.

User fairness is a key issue in power allocation for wireless NOMA systems to eliminate the near-far unfairness and ensure the performance of all users [9]. By adopting the max-min fairness rule, the authors in [10]–[12] proposed PA algorithms to maximize the minimum achievable rate in NOMA systems. In order to achieve the user fairness, more power should be allocated to the user with weaker channel. In [13], the authors designed a PA algorithm for the fairness of energy efficiency among different users in uplink NOMA system. The optimal PA factors were obtained by using Dinkelbach method. In [14] [15], low-complexity PA algorithms are proposed to maximize the proportional fairness which can help to keep balance between performance and fairness. The results of existing works validate that the user fairness aware PA algorithms can effectively avoid the performance loss of weak users. Therefore, we investigate the user fairness aware PA for NOMA assisted video transmission to guarantee the requirements for high quality of all video users.

In this paper, we focus on a downlink wireless network where two users require video streaming with the aid of NOMA. Assuming that video is processed in the unit of chunks, the video playback process is modeled by the user queue and the users can adaptively choose the video quality based on the queue condition. Furthermore, video quality degradation occurs when the user queue dose not have adequate data. The objective of this paper is to evaluate the performance of NOMA-assisted video delivery performance with adaptive quality adjustment and improve the fairness between NOMA pair from the perspective of video quality.

The main contributions of this paper are summarized as follows.

- We propose a new metric, i.e., the average video quality degradation probability, to measure the quality-of-experience (QoE) of users with the adaptive quality adjustment. Then, based on the NOMA principles, the average video quality degradation probability of each user is derived.
- To achieve the user fairness, the power allocation (PA) problem for NOMA is formulated to minimize the maximum average video quality degradation probability. Furthermore, a bisection search based PA algorithm is adopted to find an optimal solution.
- Considering the stochastic character of wireless video transmission, the theoretical analysis is capable to obtain the average video quality degradation probability. Compared with the fixed NOMA scheme, the fractional transmit power allocation (FTPA) NOMA scheme, the achievable rate fairness (ARF) NOMA scheme, and the optimized time division multiple access (TDMA) scheme, the proposed NOMA scheme is superior to achieve better user fairness from the perspective of video quality.

The rest of this paper is organized as follows. In Section II, the NOMA-assisted system model and the video playback model are respectively presented. The average video quality degradation probability is derived in Section III. A user fairness aware PA algorithm is introduced in Section IV. In Section V, simulation results and discussions are provided. Finally, the paper is concluded in Section VI.

II. System Model

In this section, we present the NOMA-assisted cooperative video transmission system, followed by the video playback model and the definition of performance metric.

A. NOMA-Assisted System

We consider a single-carrier downlink NOMA wireless system, where a source delivers different video files to two paired users, denoted by U_k, $k \in \{1,2\}$[1]. In the system, all nodes are equipped with single antenna. The video transmission period is divided into time slots with equal length t_0. It is assumed that all channels experience block Rayleigh fading. Specifically, the channel coefficient remains constant in a time slot and independent and identically distributed (i.i.d.) in different time slots. Considering the small scale and large scale fading, the channel gain in the i-th time slot is assumed to be $\lambda_{SU_k}(i) = |h_{SU_k}(i)|^2\gamma_{SU_k}$, where $h_{SU_k}(i) \sim \mathcal{CN}(0,1)$ and γ_{SU_k} is a distance-dependent constant. In this case, λ_{SU_k} follows an exponential distribution with mean γ_{SU_k}.

Considering the complexity of wireless channel, the source merely knows the partial channel state information (CSI) including the channel distribution information (CDI) resulting from the limited feedback [19]. Without loss of generality, we assume that $\gamma_{SU_1}(i) < \gamma_{SU_2}(i)$. Therefore, U_1 and U_2 refer to “the weak user” and “the strong user” of the NOMA pair, respectively. The transmission power of the source is assumed to be subject to a constraint P_0.

Let $s_k(i)$ denote the signal intended for U_k, $k \in \{1,2\}$, in the i-th time slot, where $E[|s_k(i)|^2] = 1$. The source adopts superposition coding and broadcasts $\{s_1(i), s_2(i)\}$ to both users simultaneously. The received superimposed signal of U_k in the i-th time slot can be represented by $y_k = h_{SU_k}(i)\sum_{k=1}^{2}(\sqrt{\alpha_k P_0}s_k(i)) + n_R$, where α_k denotes the the power allocation coefficient for U_k with $\alpha_1 + \alpha_2 \leq 1$ and $n_R \sim \mathcal{CN}(0,\sigma^2)$ is the additive white Gaussian noise (AWGN) at U_k. According to the principle of NOMA, more transmission power should be allocated to the weak user, i.e., $\alpha_1 > \alpha_2$. U_1 directly decodes its desired signal $s_1(i)$ by treating $s_2(i)$ as noise. As the result, the achievable rate to decode $s_1(i)$ at U_1 in the i-th time slot can be represented by

$$r_{SU_1}^{s_1}(i) = W\log_2\left(1 + \frac{\alpha_1\rho\lambda_{SU_1}(i)}{\alpha_2\rho\lambda_{SU_1}(i)+1}\right), \tag{1}$$

where $\rho = P_0/\sigma^2$. To deal with the intra-user interference within the NOMA pair, successive interference cancellation (SIC) is implemented for the decoding process of U_2 [20], [21]. Specifically, before decoding $s_2(i)$, U_2 first detects $s_1(i)$ and then removes it from the received signal. The achievable rate to decode $s_1(i)$ at U_2 in the i-th time slot can be obtained as

$$r_{SU_2}^{s_1}(i) = W\log_2\left(1 + \frac{\alpha_1\rho\lambda_{SU_2}(i)}{\alpha_2\rho\lambda_{SU_2}(i)+1}\right). \tag{2}$$

Then, $s_2(i)$ can be decoded by U_2 without the interference caused by $s_1(i)$. Therefore, the achievable rate of $s_2(i)$ at U_2 in the i-th time slot can be given by

$$r_{SU_2}^{s_2}(i) = W\log_2\left\{1 + \alpha_2\rho\lambda_{SU_2}(i)\right\}. \tag{3}$$

To successfully implement SIC, U_2 should be able to decode $s_1(i)$. Therefore, the achievable rate for U_1 and U_2 in the i-th time slot can be respectively expressed as

$$r_1(i) = \min\left\{r_{SU_1}^{s_1}(i), r_{SU_2}^{s_1}(i))\right\} \text{ and } r_2(i) = r_{SU_2}^{s_2}(i). \tag{4}$$

B. Video Playback Model

The requested video files are partitioned into many equal-size chunks to be delivered to the users in sequence [20]. In this case, the users receive and process data for video playback in terms of chunks. Specifically, chunks with size M_k bits are decoded together as a video unit to be displayed in a time frame which consists of L time slots. When the number of decoded chunks in the j-th time frame equals to $c_k(j)$, the video unit size in this time frame can be measured by $V_k(j) = c_k(j)M_k$. Here, a larger $V_k(j)$ always corresponds to a higher video quality. Assume that each user has a target required video unit size $V_k^* = c_k^* M_k$. It indicates that the target video unit should contain c_k^* chunks. Therefore, the video playback process can be modeled by the user queue. Let $Q_k(j)$ denote the queue length of U_k (in the number of target chunks) at the beginning of the j-th time frame. If $Q_k(j) \geq c_k^*$, the video can be played with target quality, i.e., $c_k(j) = c_k^*$. Otherwise, U_k adjusts the video unit size as $V_k(j) = Q_k(j)M_k$, i.e., $c_k(j) = Q_k(j)$. According to the queuing theory, the queue condition of U_k, $k \in \{1,2\}$, can be represented by the Lindley equation

$$Q_k(j) = \max\left\{Q_k(j-1) - c_k^*, 0\right\} + A_k(j), \tag{5}$$

where $A_k(j)$ denotes the number of received target chunks of U_k in the j-th time frame. Furthermore, there holds $Q_k(1) = 0$. It indicates that there is no video chunk to play in the first time frame.

According to (5), $V_k(j)$ is subject to $Q_k(j)$. The adaptive quality adjustment model is illustrate in Fig. 1. Furthermore, the chunk arrival process is related to the achievable rate for U_k, i.e., $A_k(j)$ which can be calculated by [22]

$$A_k(j) = \sum_{i=(j-1)L}^{jL}\left\lfloor\frac{r_k(i)t_0}{M_k}\right\rfloor, \tag{6}$$

where $r_k(i)$ can be obtained by using (4) and $\lfloor x \rfloor$ denotes the floor function. Therefore, the user queuing model can be equivalent to a queuing model with stochastic arrival process $A_k(j)$ and constant service rate c_k^*. The equivalent user queuing model is illustrated in Fig. 2.

C. Performance Metric

In the j-th time frame, if $Q_k(j) < c_k^*$, $k \in \{1,2\}$, U_k suffers from video quality degradation. From the perspective of queuing model, the video quality degradation occurrences when the user queue dose not

[1]Since the video transmission has strict delay requirement, the users should successfully decode the video chunks in time slots with millisecond range. For multi-user NOMA systems with successive interference cancellation (SIC), a user needs to decode the messages of other users before receiving the desirable message. In this case, the complexity of SIC process is in proportion to the number of user in the NOMA group. Therefore, implementing NOMA with all users in the video transmission systems will lead to the unacceptable decoding delay, especially for the wireless personal devices with limited communication capacity [16]. Two-user NOMA is favourable to the delay-sensitive scenarios [17]. Above all, we consider two-user NOMA to avoid the long delay caused by SIC. In practical multi-user systems, users are divided into NOMA pairs and the communication resources for different NOMA pairs are orthogonal with each other [18].

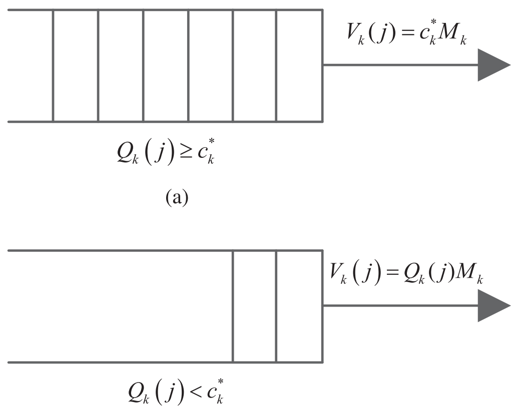

Fig. 1. The adaptive quality adjustment in the j-th time frame. (a) If the queue has adequate data, i.e., $Q_k(j) \geq c_k^*$, the video unit size is $V_k(j) = c_k^* M_k$. (b) If the queue has inadequate data, i.e., $Q_k(j) < c_k^*$, the chunk size is adaptively adjusted as $V_k(j) = Q_k(j) M_k$.

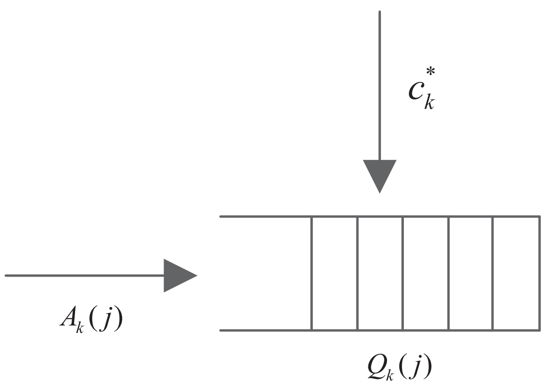

Fig. 2. The equivalent user queuing model.

have the adequate data. Define a binary matrix $\mathbf{Y} = \{y_{k,j}\}, k \in \{1, 2\}$, where $y_{k,j} = 1$ means U_k suffers from the video quality degradation in the j-th time frame and otherwise $y_{k,j} = 0$. Consequently, we have $\Pr\{y_{k,j} = 1\} = \Pr\{Q_k(j) < c_k^*\}$.

In the period $[(m-1)Lt_0, nLt_0]$, the average video quality degradation probability is defined as

$$\varepsilon_k(m,n) \triangleq E\left[\frac{\sum_{j=m}^{n} y_{k,j}}{n-m+1}\right] = \frac{\sum_{j=m}^{n} \Pr\{y_{k,j}=1\}}{n-m+1}, \quad (7)$$

where $E[\cdot]$ denotes the statistic expectation of a random variable (RV). Here, a smaller ε_k indicates that U_k experiences a better playback performance.

III. Performance Analysis

In Section II, the average video quality degradation probability is introduced as the performance metric. However, how to obtain the average video quality degradation probability is a challenge, especially considering the wireless fading channel and adaptive quality adjustment. In this section, we first analyze the video chunk arrival process for the user queue of NOMA-assisted wireless system over Rayleigh block fading channels. Then, we derive the average video quality degradation probability to measure the performance of video transmission by applying queuing theory to the equivalent user queue.

A. Video Chunk Arrival Process

As analysis above, when the target video unit size V_k^* is given, the user queue is determined by the chunk arrival process. $\lambda_{SU_k}(i)$, $k \in \{1, 2\}$, follow exponential distribution with the cumulative distribution function (CDF) $F_{\lambda_{SU_k}(i)}(x) = 1 - e^{\frac{x}{\gamma_{SU_k}}}$, for any $x \geq 0$. According to (6), for any i and $N \in \mathbb{N}^+$, the probability mass function (PMF) of $a_k(i)$ which is defined as the number of chunks received by U_k in the i-th time slot can be expressed as

$$\begin{aligned} f_k^N &\triangleq \Pr\{a_k(i) = N\} \\ &= \Pr\left\{\left\lfloor \frac{t_0 r_k(i)}{M_k^*} \right\rfloor = N\right\} \\ &= \Pr\left\{\frac{N M_k^*}{t_0} \leq r_k(i) < \frac{(N+1) M_k}{t_0}\right\} \\ &= F_{r_k(i)}\left(\frac{(N+1)M_k}{t_0}\right) - F_{r_k(i)}\left(\frac{N M_k}{t_0}\right). \end{aligned} \quad (8)$$

Furthermore, the CDF of $r_k(i)$ can be obtained as

$$F_{r_1(i)}(x) = 1 - \left(1 - F_{\lambda_1(i)}(y)\right)\left(1 - F_{\lambda_2(i)}(y)\right), \quad (9)$$

$$F_{r_2(i)}(x) = 1 - F_{\lambda_2(i)}\left(\frac{2^{\frac{x}{W}} - 1}{\alpha_2 \rho}\right), \quad (10)$$

where $y = \frac{2^{\frac{x}{W}} - 1}{\alpha_1 \rho - (2^{\frac{x}{W}} - 1)\alpha_2 \rho}$.

In the j-th time frame, considering that $a_k(i), i \in \{(j-1)L, jL\}$, are i.i.d. RVs, the PMF of $A_k(j)$ can be obtained by applying [23, eq. (17)]

$$p_k^Z \triangleq \Pr\{A_k(j) = Z\} = \sum_{\{s_0,\ldots,s_Z\} \in D_{L,Z}} \frac{L! \prod_{N=0}^{Z} (f_k^N)^{s_N}}{\prod_{N=0}^{Z} s_N!}, \quad (11)$$

where f_k^N can be obtained by applying (8) and

$$\begin{aligned} &D_{L,Z} \\ &= \left\{(s_0, \ldots, s_Z) \middle| \sum_{N=0}^{Z} s_N = L, \sum_{N=0}^{Z} N s_N = Z, s_N \in \mathbb{N}_0\right\}. \end{aligned}$$

B. Average Video Quality Degradation Probability

The average video quality degradation probability is determined by the value of $\Pr\{y_{k,j} = 1\}$. Considering $j \in [2, n]$, the queue length of U_k can be obtained by conducting iterative operation for the right-hand side of (5)

$$\begin{aligned} &Q_k(j) \\ &= \max_{1 \leq m \leq j-2} \{A_k(m, j-2) - c_k^* \cdot (j-m)\} + A_k(j-1) \\ &= \max_{1 \leq m \leq j-2} \{G_k(m, j)\} + A_k(j-1), \end{aligned} \quad (12)$$

where $G_k(m,j) \triangleq A_k(m, j-2) - c_k^* \cdot (j-m)$. According to (12), we have

$$\begin{aligned} &\Pr\{y_{k,j} = 1\} \\ &= \Pr\left\{\max_{1 \leq m \leq j-2} \{G_k(m,j)\} < c_k^* - A_k(j-1)\right\} \\ &= \Pr\{G_k(m,j) < c_k^* - A_k(j-1), 1 \leq m \leq j-2\}. \end{aligned} \quad (13)$$

Here, $G_k(m,j)$ are non-independent discrete RVs and the distribution of $G_k(m+1,j)$ is only correlated with $G_k(m,j)$.

Clearly, when $A_k(j-1) \geq c_k^*$, there holds $\Pr\{y_{k,j}=1\}=0$. For $A_k(j-1) < c_k^*$, let $S_k(m)$ denote the sample space of $A_k(m)$ to satisfy (13). Then, we have

$$S_k(m) = \begin{cases} [-c_k^*, b_k(j)], & m = j-2, \\ [f_k(m,i), b_k - G_k(m+1,j)], & \text{otherwise}, \end{cases} \tag{14}$$

where $b_k(j) = c_k^* - A_k(j-1) - 1$ and $f_k(m,i) = -c_k^* + G_k(m+1,i)$.

Above all, (13) can be rewritten as

$$\Pr\{y_{k,j}=1\} = \sum_{Z_{j-2}} \sum_{Z_{j-3}} \cdots \sum_{Z_1} \prod_{1\leq m\leq j-2} f_k^{Z_m}, \tag{15}$$

where Z_m denotes the element of $S_k(m)$, i.e., $Z_m \in S_k(m)$. Inserting (15) into (7), the average video quality degradation probability can be obtained.

IV. User Fairness Based Power Allocation

In NOMA systems, it is important to achieve fairness among different users by adopting effective PA algorithm. To balance the video quality between U_1 and U_2, the average video quality degradation probability obtained in Section III is used as the metric for PA algorithm design. In this case, the PA for the NOMA-assisted cooperative system with user fairness can be formulated as the following min-max problem.

$$\min_{\alpha_1,\alpha_2} \quad \max\{\varepsilon_1(m,n), \varepsilon_2(m,n)\}, \quad m < n \tag{16a}$$

$$\text{s.t.} \quad \alpha_1 + \alpha_2 \leq 1, \tag{16b}$$

$$\alpha_1 > \alpha_2, \tag{16c}$$

where $\varepsilon_1(m,n)$ and $\varepsilon_2(m,n)$ can be obtained by using (15); (16b) is the transmission power constraint; (16c) is based on the NOMA principle to allocate more power to the weak user.

In order to solve the min-max problem, the following proposition is proposed to find the properties of the corresponding optimal solution.

Proposition 1: Considering the min-max problem formulated in (16a)-(16c), the corresponding optimal solution $\{\alpha_1^*, \alpha_2^*\}$ should satisfy that

$$\alpha_1^* + \alpha_2^* = 1, \tag{17}$$

and ensure

$$\varepsilon_1(m,n) = \varepsilon_2(m,n). \tag{18}$$

Furthermore, $\{\alpha_1^*, \alpha_2^*\}$ is the unique optimal solution.

Proof: Please refer to Appendix A. ■

According to Proposition 1, we adopt the bisection search method in [24] to obtain the optimal solution of the min-max problem. Since the object of the PA is to make $\varepsilon_1(m,n) = \varepsilon_2(m,n)$, $|\frac{\varepsilon_1(m,n)}{\varepsilon_2(m,n)}|$ can be regarded as the metric for bisection search. If $|\frac{\varepsilon_1(m,n)}{\varepsilon_2(m,n)} - 1| > \eta$, where η is a predefined threshold, the search will continue. When η is set to be small enough, the solution obtained by using bisection search method is very close to the global optimal solution.

V. Simulation Results and Discussions

In this section, Monte Carlo simulations are conducted to evaluate the video quality of the NOMA-assisted wireless system with the proposed user fairness based PA algorithm. To validate the efficiency of the proposed NOMA scheme, we also give the results of the following comparison schemes.

- Optimized TDMA scheme, in which the optimal time sharing factor [25] can be obtained by applying a method similar to Algorithm 1.
- Fixed NOMA scheme, in which the PA coefficients are fixed to $\alpha_1 = 11/12$ and $\alpha_2 = 1/12$ for all network configurations.
- FTPA NOMA scheme, in which we have $\alpha_k = \frac{(\lambda_{SU_k})^{-\beta}}{(\lambda_{SU_k})^{-\beta} + (\lambda_{SU_{\bar{k}}})^{-\beta}}$, $k \cup \bar{k} = \{1,2\}$, where β is the fractional parameter [26].
- ARF NOMA scheme, in which the power allocation is designed to maximize the minimum achievable rate [10]–[12]. Considering the stochastic character of wireless video transmission, the power allocation for ARF NOMA can be given by

$$\min_{\alpha_1,\alpha_2} \quad \max\{E[r_1(i)], E[r_2(i)]\} \tag{19a}$$

$$\text{s.t.} \quad \alpha_1 + \alpha_2 \leq 1, \tag{19b}$$

$$\alpha_1 > \alpha_2. \tag{19c}$$

TABLE I
Simulation Parameters

Parameters	Value
Number of slots in each time frame (L)	5
Length of each time slot (t_0)	2ms
Bandwidth (W)	100MHz
AWGN power spectral density (σ^2)	-169dB/Hz
Transmission power constraint (P_0)	30dBm
Target number of chunks in each video unit (c_k^*)	25
Distance between the source and U_2 (D_2)	0.1km
Predefined threshold for bisection search (η)	10^{-5}
Fractional parameter for FTPA NOMA (β)	3

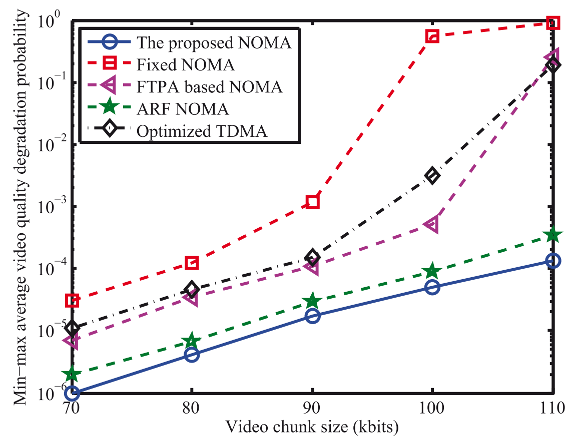

Fig. 3. Min-max video quality degradation probability versus video chunk size for different transmission schemes with D_1=0.4 km.

In the simulation, we consider the video files consist of units with mutually exclusive information for the playback process in a time frame. In [27], the video unit is termed as group of picture (GOP). If all chunks for a video unit can be successfully decoded by the user, the video can be displayed with the target quality. Otherwise, the quality should be adjusted based on the user queue condition, which will cause the performance degradation. For transmission model, we consider a video transmission period with 2×10^3 time frames. Furthermore, the distance-dependent path loss is modeled as $128.1 + 37.6\log_{10}(D_k)$, where D_k in kilometer is the distance between the source and U_k, $k \in \{1,2\}$. The simulation parameters are listed in Table I.

Figs. 3 and 4 illustrate the min-max video quality degradation probability comparison of different transmission schemes for the varying video chunk size, i.e., M_k, and distance between the source and U_1, i.e., D_1, respectively. Clearly, we can see that the proposed NOMA

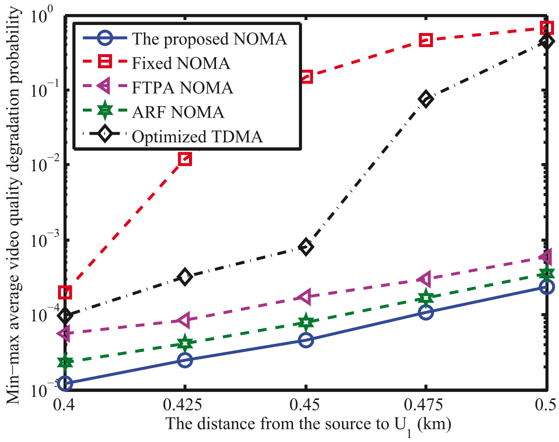

Fig. 4. Min-max video quality degradation probability versus the distance between the source and U_1 for different transmission schemes with $M_k = 80$kbits.

scheme performs best. By adopting the proposed NOMA scheme, the video quality of U_1 can be improved at the cost of video quality degradation of U_2. According to Proposition 1, the PA algorithm in the proposed NOMA scheme can maximize the overall user fairness from the perspective of the video quality. Furthermore, the min-max video quality degradation probability of the proposed NOMA scheme can be maintained within an acceptable range with different network configurations. It indicates that the quality requirements of both users can be satisfied.

The fixed NOMA scheme is based on the channel independent PA algorithm, where the channel coefficients are merely subject to $\alpha_1 > \alpha_2$. Although FTPA can dynamically adjust the PA based on the channel conditions, it cannot completely eliminate the near-far unfairness between the NOMA pair. ARF NOMA can achieve the user fairness in a time frame. However, the overall user fairness can not be guaranteed. Furthermore, compared with the optimized TDMA scheme which is also designed to achieve user fairness, the proposed NOMA scheme can provide higher spectrum efficiency. Based on the results, why the proposed NOMA scheme outperforms the fixed NOMA scheme and the optimized TDMA scheme can be explained. Besides, we see that the fixed NOMA is inferior to the optimized TDMA in video quality guarantee. It indicates that the performance of NOMA is highly depended on the PA algorithm.

VI. Conclusion

In this paper, we analyze the average video quality degradation probability of a NOMA assisted wireless system consisting of a NOMA pair. Video file is divided into equal-size chunks to be processed. The chunk deliver rate of each user is obtained based on the principle of NOMA with SIC. Then, the average video quality degradation probability is derived to evaluate the performance. A bisection search algorithm is adopted to solve the PA problem which is formulated as a min-max problem. Simulation results demonstrate the advantage of the proposed NOMA scheme in achieving the fairness between two users compared with the existing schemes.

Appendix
Proof of Proposition 1

We first prove that (18) is the necessary and sufficient condition for the optimal solution with $\alpha_1 + \alpha_2 = 1$. For $\{\alpha_1^*, \alpha_2^*\}$, the corresponding average video quality degradation probability is denoted by $\varepsilon_1^*(m,n) = \varepsilon_2^*(m,n) = \varepsilon^*(m,n)$. When $\alpha_1' = \alpha_1^* + \delta$, where $\delta > \frac{\alpha_2^* - \alpha_1^*}{2}$, we have $\alpha_2' = \alpha_2^* - \delta$ with corresponding average video quality degradation probability $\{\varepsilon_1'(m,n), \varepsilon_2'(m,n)\}$. According to (1) to (4), one can easily prove that when the channel coefficients are given, the values of $r_{S_1}^{s_1}(i)$ and $r_{SU_2}^{s_1}(i)$ are monotone increasing with α_1. It indicates that the relationship between $\varepsilon_1(m,n)$ and α_1 is always negative. For $\varepsilon_2(m,n)$ and α_2, the similar results can be obtained. Consequently, there holds

$$\begin{cases} \varepsilon_1'(m,n) < \varepsilon^*(m,n), \varepsilon_2'(m,n) > \varepsilon^*(m,n), & \delta > 0, \\ \varepsilon_1'(m,n) > \varepsilon^*(m,n), \varepsilon_2'(m,n) < \varepsilon^*(m,n), & \text{otherwise}. \end{cases} \tag{20}$$

According to (20), we have $\max\{\varepsilon_1'(m,n), \varepsilon_2'(m,n)\} > \varepsilon^*(m,n)$. Above all, we see that $\{\alpha_1^*, \alpha_2^*\}$ is the unique feasible PA coefficient to min-max $\{\varepsilon_1(m,n), \varepsilon_2(m,n)\}$ among all solution with $\alpha_1 + \alpha_2 = 1$.

Then, we prove that $\{\alpha_1^*, \alpha_2^*\}$ is the optimal solution of the min-max problem formulated in (16a)–(16c). For any solution $\{\alpha_1', \alpha_2'\}$, where $\alpha_1' + \alpha_2' < 1$, if $\alpha_1' < \alpha_1^*$, there holds $\varepsilon_1'(m,n) > \varepsilon_1^*(m,n)$. Otherwise, we have $\alpha_2' < \alpha_2^*$, which indicates $\varepsilon_2'(m,n) > \varepsilon_2^*(m,n)$. Therefore, we have $\max\{\varepsilon_1'(m,n), \varepsilon_2'(m,n)\} > \varepsilon^*(m,n)$. Here, the proof of Proposition 1 is completed.

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